Tech Lead Take-Home Exercise: Async Credit Processing Service

Overview

Build a small asynchronous service that credits a user's balance through a FastAPI endpoint, a message queue, and a Celery worker backed by MongoDB.

The Problem

A client calls an API to add credit to a user's account; the credit is applied asynchronously, and the record shows its progress through a status field.

1. The client sends `POST /credits` with a `user_id` and an `amount`.
2. The FastAPI endpoint validates the request, sets the user's MongoDB record `status` to `processing`, and publishes a task to the queue.
3. The endpoint returns immediately (for example, `202 Accepted`) with a task ID.
4. A Celery worker picks the task from the queue and adds the `amount` to the user's `balance` field (creating the field if it does not exist).
5. The worker sets the `status` to `processed` once the balance is updated.

Example user document after processing:

```
{ "user_id": "u_123", "balance": 150.00, "status": "processed" }
```

Requirements

Use Python 3.11+, FastAPI, Celery, MongoDB, and a message broker of your choice. Build it like production code your team ships and maintains.

Deliverables

1. **GitHub Repository:** Submit a link to a GitHub repository containing the complete source code and a `README.md` file with clear instructions for reviewers to install, run, and test your solution.
2. **Two-Page Design Document:** Include a separate, two-page design document describing the solution's architecture, key design decisions, implementation approach, and testing strategy. The document should explain the major components, how they interact, and the rationale behind the chosen approach.
3. **Loom Video Recording:** Submit a link to a Loom video (approximately 5–10 minutes) walking through your solution (other video tools are ok, just make sure we can view it!). The recording should demonstrate the application running, explain the key architectural and implementation decisions, highlight important aspects of the code, and show relevant tests. Briefly discuss any trade-offs, assumptions, or areas for future improvement.

Submission

Reply with your deliverables within 5 days of receiving this exercise. Expect a 45-minute follow-up call to walk through your solution. Questions are welcome at any time.